INCH-POUND

A-A-58063
15 June 1996

COMMERCIAL ITEM DESCRIPTION

High Speed Snow Broom, Mounted On A 44,500 LBS GVW (minimum), 6x4 Dump Truck

This description covers a diesel-engine-powered push type high speed rotary sweeper with snow blower that is mounted on a 44,500 LBS GVW (minimum), 6x4 dump truck. The rotary broom shall be mounted on the vehicle's front, a skid mounted diesel engine shall be in the dump body and the blower shall be mounted to the rear of the vehicle.

Salient Characteristics:

A. Brush assembly.

- cylindrically shaped cone

- mounting frame

- hood and deflector

- angling system

- drive system

- castors

- hitch

- full length of core shall be filled w/flat ring, continuous wafer type fillers

- spacers (steel or plastic)

- adjustable, to provide different sweeping patterns by using mechanical adjuster that locks into place when released

- no tools required to make sweeping pattern adjustments

B. Brush core.

- two (2) cores, 8 ft (min) each section

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: WR-ALC/LVRE, 225 Ocmulgee Ct, Robins AFB, GA 31098-1647.

AMSC N/A FSC 3825

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- 46" +0 -1/16" diameter

- replaceable bristles

- middle of brush supported by bearings

- replaceable filler lug channels

C. Brush frame.

- Welded steel construction, frame components may be removable to allow for ease of brush replacement

D. Brush hood.

- cover 40% (min) of brush circumference

- 14 gauge (min) steel

- steel hinged

- welded or bolted to brush frame

- provisions for mounting snow deflector on hood front

- non clog design

- provide quick access to brush for replacement of brush sections and inspections

- weight support "stops" when open

E. Hood air deflector and director.

- metal flap mounted on front edge of hood to change angle of swept snow leaving the brush

- the flap shall be hydraulically controlled from cab

F. Brush angularity system.

- 45 degrees right or left from transverse position capability (the brush shall sweep the same path whether angled right or left)

- selectable angling capability, hydraulically actuated from cab

- self leveling brush head

- "stops" to prevent brush contact with brush push frame

G. Brush drive system.

- brush core, driven by closed loop hydrostatic system

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- 0 - 500 rpm

- drive motor(s) shall be protected within the brush framework

H. Castering system.

- pneumatic tired, 16 ply min., nitrogen - filled 18x7x8, 3600 revolving, single type castors

- shimmy type dampeners

- at least 70% of broom assembly weight shall be supported by 4 castors spaced along the frame

- castors to track within sweeping path at all times

I. Quick hitch.

- quick attachment and removal capability by means of quick hitch device

- brush assembly support when not attached, the support shall maintain the broom in approximately the same inclination as when installed

J. Air blower.

- double outlet centrifugal type

- fan capacity, 10, 000 ft/min, minimum

- air velocity, 325 mph, minimum, at each blower outlet

- blower fan, minimum range of 25 - 2600 rpm

- air blast to automatically change when brush angle changes

- air blast to be perpendicular to direction of travel and toward direction of brush discharge

- air discharge nozzles shall have manual adjustment directing air blast to outside sweeper unit for clearing run way lights

- air discharge nozzles, capable of vertical movement, allowing runway lights to be cleaned, top to bottom

K. Blower mounting.

- adjustable quick disconnect type, attached at rear of engine skid

- fork lift tine slots, front and rear, to prevent dropping the assembly

L. Engine skid assembly.

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- diesel engine, continuous hp rating of 190 hp (in accordance with SAE J1349), water cooled, the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions

- metal skid mounted, capable of fitting and being secured in bed of 44,500 GVW (min), 6x4 dump truck, removable by crane or fork lift

- two forklift "C" type tine channels

- double hydrostatic pump drive mounted to engine's flywheel housing for direct drive of the brush and blower pumps

- automatic high temperature and low oil cut off devices

- electrical starting system

- instrumentation (to include but not limited to the following)

  - engine hour meter

  - ammeter

  - oil pressure gauge

  - fuel gauge

  - coolant temperature indicator

  - turbo type precleaner in air cleaner system

  - fuel tank, 8 hours operating capacity

  - electromagnetic radiation limited IAW SAE J551

M. Cowling.

- louvered sheet metal cowling, 16 gauge minimum, for engine skid assembly and hydraulic components (slots in the panel over skid engine radiator intake may be required if necessary to meet air flow requirements for the radiator)

- hinged access side panels

- removable with common hand tools

N. Hydraulic system.

- necessary components, valves and plumbing

- broom and blower drive circuits

- hydraulically actuate sweeper components

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- automatic temperature control bypass

- reservoir with baffles , 15 gal minimum, with outside changeable filtering system

- oil sight and oil temperature gauges

- 8 micron absolute full flow bypass type filter on high pressure side

- 10 micron absolute, full flow, no bypass filter on low pressure side

- restriction gauges or devices to indicate replacement required

- use ISOVG 46 hydraulic oil

O. Hoses, tubing and coupling.

- correct length umbilical cord from pump to brush frame, routed/supported to prevent excessive rubbing against the truck

- high pressure hose, 4 or 6 spiral wire

- rated working pressure, 5000 psi minimum

- hoses and tubing, equipped with male and female quick couplers, fool proof design

P. Lighting.

- two auxiliary driving lights on broom push frame

- connected with electrical quick disconnect

- two auxiliary tail/stop/signal lights and marker lights on rear of blower or skid assembly with hardware allowing connection to truck lighting system

- minimum two working lights for engine area

- warning beacon, visible from 360 degrees, mounted on engine skid assembly, 90 flashes/min (beacon shall be class 2 flashing type lamp in accordance with SAE J845 with yellow globe conforming to SAE J578)

- all controls on the control box

Q. Battery.

- sealed maintenance free type

- located for easy access, protected from the weather by a battery box or engine cowling

R. Brush and blower remote controls.

- umbilical cord to control box in cab

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- quick disconnect coupler multipin type, with hinged weatherproof cover to be permanently mounted through right rear corner of dump truck

- gauges, controls and switches to be logically arranged and grouped

- instrumentation

  - hydrostatic motor fluid pressure gauge

  - hydraulic fluid temperature gauge

  - auxiliary engine brush and blower speed gauges

  - engine start switch with key

  - engine oil pressure warning light

  - engine coolant temperature warning light

  - speed controls for engine, brush and blower

  - control for raising/lowering brush

  - control for angling brush

  - control for brush hood air deflector position

  - manual switch for vertical oscillation of blower ducts

  - switches for warning beacon, auxiliary rear lights, auxiliary driving lights and engine skid assembly work lights

  - lighted control box

S. Truck servicing shall not be interfered with when complete assembly is mounted on truck.

T. Hydraulic system sensors.

- compatible with UCC monitor, part number US-S20M-6526411111

- installed in both brush and blower drive circuits

U. Painting.

- Treated, primed and painted (acrylic urethane paint) IAW manufacture's standard commercial practice

V. Identification marking.

- An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

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nomenclature
manufacturer's model number
manufacturer's serial number
manufacturer's name
manufacturer's address
manufacturer's service phone number
date of manufacture
contract number
national stock number
registration number

W. Lubrication.

- moving parts and pivot points shall be furnished with replaceable bearings, bushings or wear surfaces, all shall have provisions for grease lubrication

- lubrication fittings, IAW SAE J534

- decal, securely attached

- point of application

- service interval

- type of lubrication

- viscosity

X. Special tools.

- manufacture supplied if required

Y. Temperature conditions.

- crank and operate, -20^{0}F

- measured shot ether injection system

- storage temperature, -65^{0}F

Z. Safety.

- moving parts and parts subject to high temperature shall be guarded or insulated

- warning/caution markers to identify potential hazards

- ladder, to assist entering dump bed

- hydraulic component burst pressure shall be 150% of rated working pressure for the system

- engine skid and blower assemblies shall be secured to prevent shifting in dump body

AA. Dump truck.

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- 44,500 LBS GVW (minimum), 6x4 IAW Fed-Std-807A (or latest revision) Item No. 643B

- The engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions.

BB. The broom/blower/engine skid furnished must be the latest model as of the date of award (with minor modifications, as that term is defined in DoD FAR Supplement 252.211-7012, Paragraph K-85 of the solicitation) of an established product previously produced and sold in substantial commercial quantities (exceeding the criteria at FAR 15.804-3(f)(2)(ii)(A)) for at least three years immediately preceding the issuance date of the solicitation. The broom/blower/engine skid furnished, or earlier models of it, absent minor modifications, shall have routinely been supported with spare/repair parts which were produced or sold in the normal course of business. At the time of delivery, the contractor shall furnish the Administrative Contracting Officer with verification of compliance with these requirements.

NOTE TO OFFERORS: Offerors shall provide, as part of their proposals, the following information on the broom/blower/engine skid (or its earlier models) being offered as a commercial item:

(1) Total sales to the U.S. Government or to contractors for U.S. Government use during the three years immediately preceding the issuance date of the solicitation.

(2) Total sales of the item to the general public during the three years immediately preceding the issuance date of the solicitation.

(3) Length of time the item has been sold in the commercial market place.

(4) Points of contact and phone numbers of customers to which the item has been sold in the commercial market place.

(5) Other pertinent information necessary to determine that the item is an established commercial item.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

NOTES: When additional hitches are required for particular trucks the locations of the trucks shall be furnished to the contractor. It will be the contractor’s responsibility to obtain the information (measurements etc.) necessary to fabricate the required hitches.

Custodian:
AIR FORCE-99

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